Paper 01:

Title: Study of the performance of Deep learning methods used to predict tidal current movement.

Summary:

The main goal of this research was to accurately predict tidal current movement in the ocean, so that electricity can be generated more efficiently from the sea. First, a computer software called **ROMS** was used to calculate ocean current velocity but this calculation always contains some errors, because the computer cannot perfectly capture every real-world factor like wind and complex coastlines. At the same time, a device called **ADCP** was placed in the Zhoushan sea in China to measure the actual real-world current every 10 minutes for about 40 days. Then these two datasets were used together to train a **Deep Learning Model** the ROMS simulated data was given as Input, and the ADCP measured data was given as the correct answer. The model tried again and again, each time reducing its mistakes, until it learned the relationship between what ROMS says and what actually happens in reality. Once training was complete, ADCP data was no longer needed simply giving new ROMS data to the trained model was enough to get a corrected, accurate result. In this research, three different models **MLP, LSTM and AR-ANN** were tested under the same conditions, and it was found that AR-ANN performed the best, reducing the original simulation error by approximately **42%**. Finally, this corrected current data was used to calculate how much tidal energy could potentially be generated at that location, giving a much more accurate assessment than before.

Abstract:

This paper combines 2 processes:

No.01: Deep learning method: for comparative investigation using models: MLP, LSTM, AR-ANN (hybrid model)

No.02: numerical simulation- using ROMS (Regional Ocean Modeling System) software.

Using AI, the term ROMS most commonly refers to the Regional Ocean Modeling System, an advanced oceanographic modeling software used by scientists worldwide. However, depending on the context, it may also refer to Read-Only Memory in computing or ROM images used in gaming.

Using this we can simulate problem base on mathematical calculation.

a. Compared with the numerical simulations, deep learning methods can increase the original correlation coefficient from 0.4 to over 0.8
b. The AR-RNN model shows excellent performance both meridional and zonal
c. This advantage of deep learning algorithms shows MLP, LSTM and AR-ANN models reducing the root mean square error by 32.9%, 34.4% and 42%, respectively.
d. The new method can be used to accurately predict the hydrodynamic of tidal flow in the process of tidal energy extraction, which contributes to determine the suitable location for energy generation and tidal turbine design.

Introduction:

The existing methods for tidal current prediction are 4 types:
statistical, dynamical (numerical), artificial intelligence, and hybrid models.

No. 01: Statistical methods (such as harmonic analysis) predicted tidal currents by representing them as a combination of several sine waves. However, these methods had low accuracy because they ignored many real-world factors such as wind, turbulence, and seabed conditions.

No. 02: Dynamical or numerical models (such as ROMS) were used, which apply physics-based equations to simulate tidal behavior. Although these models provide better results, there is still a deviation from real-world data.

No. 03: Artificial intelligence (AI) and deep learning methods (such as MLP and LSTM) were introduced, which learn patterns from data and provide more accurate predictions, especially for time-series data.

No. 04: The most powerful approach is the hybrid model, where numerical simulation and deep learning are combined to detect and correct simulation errors, resulting in significantly improved accuracy in tidal current and tidal energy prediction.

In this paper they propose a novel neural network, namely the attention-ResNet neural network (AR ANN). By adding an attention mechanism to the ANN, this method attempts to improve the deviation of the tidal velocity in amplitude, which greatly affects the accurate calculation of the tidal energy. Therefore, the work in this paper will conduct comparative experiments on these three neural networks to find the optimal model while exploring the respective mechanisms.

Methods:

Here this tidal current prediction model showed in this paper mainly of three parts

1. To predict and analyze tidal current changes in target seas using a numerical model; ROMS show a simulation of tidal prediction and provide a minimum guarantee.
2. To construct neural networks based on simulation results and measurement data using different deep learning methods;
3. To calibrate tidal current velocity for non-observed periods using the neural networks above.

Datasets

1. **Data Collection**
   - Location: **Zhoushan, China** (Lat: 30.116° N, Lon: 122.690° E)
   - Device: **Acoustic Doppler Current Profiler (ADCP)** measures water current velocity using sound waves
   - Measurement period: **30 May – 9 July 2021**
   - Sampling interval: **every 10 minutes**
2. **Data Quality Control**
   - Remove obvious outliers (sudden changes)
   - Fill missing data using **smoothing methods**
   - Velocity split into **zonal (u)** and **meridional (v)** components
3. **Data Preparation for Neural Networks**
   - Sequence length: **12 consecutive hours = 72 data points per sample** (**Sequence length:** 12 hours × 6 readings/hour = **72 data points per sample)**
   - Total samples: **5700(**40 days × 24 hours × 6 readings/hour = **5,760 readings**
     - Training: 1-4400

        - Validation: 4400-4700
        - Testing: 4700-5700
4. **Neural Network Training**
    - **Input →** ROMS simulated current data (computer-generated)
    - **Output →** ADCP measured current data (real-world observed)
    - Networks: **MLP, LSTM, and AR-ANN**
    - Hyperparameter tuning: exhaustive search
        - Dense/LSTM units: 100, 200, 300, 400
        - Number of heads: 1–10
    - Optimizer: **Adam**, Loss: **MSE**
    - **Early stopping** applied with patience = 20
5. **Purpose**
    - To optimize model performance
    - Compare results across different neural network types

Table 1. Detailed information for deep learning algorithms.

Parameter	MLP	LSTM	AR-ANN
Input Shape	(None,72)	(72,1)	(None,72)
Batch Size	100	100	100
LSTM Units	/	300/200/200	/
Dense Units	300/200/200	/	300/200
Head Numbers	/	/	6
Dropout Rate	0.3	0.3	0.3
Epoch	100	100	100

Table 2. Error metric for tidal current speed u.

Variable	Metric	ROMS	MLP	LSTM	AR-ANN
U	RMSE	0.234	0.160	0.150	0.146
	MAE	0.200	0.143	0.121	0.113
	R	0.484	0.811	0.820	0.830
V	RMSE	0.176	0.163	0.161	0.153
	MAE	0.173	0.153	0.147	0.152
	R	0.814	0.848	0.866	0.892

4. Tidal Current Energy Assessment

The proposed model was applied to the tidal energy assessment, and its effectiveness was verified. The power density is commonly used to measure the tidal energy, and its magnitude is mainly limited by the speed of the tidal current. The specific formula is as follows:

$$P = \frac{1}{2}\rho(\sqrt{u^2 + v^2})^3 \tag{21}$$

where P is the tidal power density and ρ is the density of seawater, and the value is taken as 1025 kg/m^3.

Detailed comparison results (RMSE, MAE, and R evaluation) of the tidal current energy assessment are illustrated in Table 3. In terms of the RMSE, the values of the three models were 50.096, 49.003, and 43.315, respectively. Compared to the original value of 74.68, the three methods reduced the error by 32.9%, 34.4%, and 42%, respectively. In the evaluation of the tidal energy, the deep learning methods had advantages in evaluating the tidal energy and showed superior results.

Table 3. Error metric for tidal current energy.

Metric	ROMS	MLP	LSTM	AR-ANN
RMSE	74.68	50.096	49.003	43.315
MAE	67.467	43.834	39.505	38.389
R	0.297	0.714	0.743	0.786

Summary Table

	Previous Work	This Paper
Method	ROMS alone or DL alone	ROMS + Deep Learning combined
Neural Network	MLP or LSTM only	New AR-ANN proposed
Focus	Velocity prediction only	Velocity + Energy assessment
Accuracy improvement	Limited	Up to 42% error reduction

This paper introduced a **new hybrid framework** combining ROMS simulation with a **newly designed AR-ANN** model to not only predict tidal currents more accurately but also improve tidal **energy assessment** none of which had been done together before.